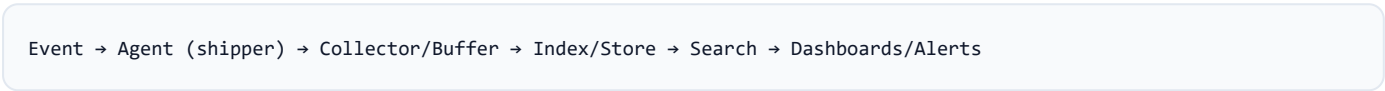


SIEM Fundamentals & Searching Logs

The SIEM Pipeline



Every login, process start, DNS query, and file change that ends up searchable went through this pipeline. The SIEM is the **memory of the network** – and memory is what you analyze when something goes wrong.

The Log-Types Agenda

What a SOC lives on:

Source	Class of events	Example threats visible
Authentication	Logon/Logoff	brute force, impossible travel
Endpoint (EDR)	Process, file, network	malware, suspicious child processes
Network	Firewall/DNS/Proxy	C2 beacons, data exfil
Cloud/cloud	IAM, storage, API	privilege changes, data download spikes
Web/DB	HTTP, SQL	SQLi, API abuse

Searching: The Language of a SIEM

Elastic/ELK (KQL):

event.category:authentication AND event.outcome:success
failures within window: event.action:"logon failure" AND user.name:"svc-"

SpLunk (SPL):

```
index=windows EventCode=4625 | stats count by src_ip | sort -count
```

Common queries you'll write immediately:

`4625` (failed login) grouped by source
new account creation (`4720`)
unusual PowerShell execution (`Image: powershell.exe` from odd parent)
outbound connections from a server to the internet

The "Fifth Question" Discipline

Every investigation answers:

Who – accounts, hosts

What – actions, tools

When – timestamps, windows

Where – sources, destinations

Why does this matter? – asset value, blast radius

Lab Fast-Start (Elastic/OpenSearch, free)

```
# easiest: run the official Elastic docker-compose (or use cloud free tier)
# then configure a Filebeat agent on one Linux VM:
#   filebeat modules enable system
#   filebeat setup; filebeat -e
# now search "system.auth" events in Discover
```

The skill is query fluency. The more log types you can instantly write sightings to, the faster your MTTD shrinks. Practice 30 minutes/day of query writing on your own lab.

Normalization & the ECS

Different vendors name the same event differently ("user", "account", "winlog.user"). The **Elastic Common Schema (ECS)** or SIEM's own normalizer translate them into one vocabulary – so an analyst searches one schema, not fifty.

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